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RESEARCH REVIEW

SMART HOSPITAL MANAGEMENT SYSTEM

A Literature Review of Hospital Information Systems, Security, Usability and Digital Workflow Management

Submitted by

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ABSTRACT

Hospital Information Systems (HIS) support the administrative and information-management activities of modern hospitals. Their purpose is not limited to storing patient records; they can also coordinate appointments, staff, pharmacy, laboratory work, billing, admissions, reporting and communication. However, a hospital system is useful only when its information is accurate, its workflows are understandable, access is controlled and the software remains dependable during routine use. This review examines published research on hospital information-system evaluation, technical requirements and electronic-record usability, and connects those findings to the design of SmartCare, a student-developed Smart Hospital Management System.

The reviewed literature shows that HIS evaluation is multidimensional. Khorasani et al. (2026) grouped evaluation indicators into human, technological, informational, organizational, legal and patient-safety domains. Farzandipour et al. (2020) proposed a technical-requirements framework covering communication services, system architecture, security and response time. Wronikowska et al. (2021) found variation in the usability methods and metrics used to evaluate hospital electronic-record systems. Together, these studies suggest that a system should be evaluated through both technical checks and user-centred workflow testing.

This document is a literature review prepared to support the academic project; it is not an original clinical study or a peer-reviewed research article. It does not claim that SmartCare has been clinically validated. The project-specific discussion translates published findings into design considerations and a proposed evaluation plan. Actual performance, security and usability results must be established through documented testing.

Keywords: Hospital Information System, Smart Hospital Management System, Electronic Health Records, Usability, Data Security, Appointment Management, Interoperability, Java, MySQL.

DOCUMENT STRUCTURE

The review is organised into background and objectives, literature-search approach, thematic analysis of research, relevance to SmartCare, a proposed test and evaluation framework, limitations, conclusion and references. Published findings are distinguished from recommendations made for the student project.

1. INTRODUCTION

Hospitals handle many related processes at the same time. Patient registration, appointment scheduling, consultation, diagnostic testing, medicine dispensing, admission, billing and discharge may involve different departments and staff members. When these activities are recorded in disconnected files or paper registers, staff can face duplicated entry, delays in locating information, inconsistent records and difficulty preparing management reports. A Hospital Information System aims to organise such information and support coordinated workflows.

A digital system does not automatically guarantee better care. Poorly designed screens can slow users down; weak access controls can expose sensitive information; inconsistent data can create confusion; and unreliable integrations can prevent departments from sharing updates. Therefore, evaluation should consider more than whether a page opens or a form saves. It should examine the accuracy, security, usability, reliability and organisational fit of the system.

SmartCare is considered in this report as a software engineering project that brings several common hospital-administration modules into one application. Its described technology stack includes Java 17, Jakarta Servlet/JSP, JDBC, MySQL, Maven WAR packaging, BCrypt and Docker. The module scope includes authentication and administration, patient and doctor records, appointments and consultation, pharmacy, laboratory, nursing, billing, admissions, reports and feedback. These project details define the context for applying research findings; they are not evidence that each workflow has passed independent testing.

The review has three objectives:

1. Summarise research findings that are relevant to hospital information-system quality.
2. Identify design and evaluation considerations for SmartCare, particularly usability, security, data quality and integration.
3. Propose a practical testing framework that can be used to assess the project without claiming unmeasured outcomes.

The central question is: *Which quality factors from published HIS research should guide the design and evaluation of a student-developed Smart Hospital Management System?*

2. BACKGROUND AND CONCEPTS

2.1 Hospital Information Systems

An HIS is an organised combination of software, hardware, data, users and operating procedures used to manage hospital information. It may support clinical records, administration, finance, pharmacy, laboratory services, scheduling and reporting. The exact boundaries differ between hospitals: one organisation may use a single integrated platform, while another may connect several specialised applications.

2.2 Electronic Health Records and Administrative Records

Electronic health records (EHRs) organise information related to an individual’s care over time. Administrative records support activities such as registration, appointment allocation, admission and billing. These information types are related but should not be treated as identical. Access to clinical information should be appropriate to a user's responsibilities, and administrative access should not automatically imply unrestricted access to every clinical detail.

2.3 Workflow Integration

Integration means that related steps and data remain consistent across modules. For example, an appointment may refer to a registered patient and a doctor; a consultation may be associated with that appointment; and a bill may contain charges related to services actually recorded. Integration is not simply placing links to different pages in a common menu. It also involves data relationships, validation rules, error handling and traceable updates.

2.4 Quality Attributes

The quality of a hospital system can be discussed through functional suitability, usability, reliability, performance, security, maintainability and compatibility. In a healthcare context, informational quality and patient safety are especially important because mistakes in records or confusing workflows can have consequences beyond ordinary office software. A student project can demonstrate software-engineering principles, but it should not be presented as clinically safe without appropriate validation and governance.

2.5 Scope of This Review

The report focuses on system evaluation and implementation considerations rather than medical diagnosis or treatment algorithms. It uses published studies to discuss software and organisational quality. SmartCare is used as an application context for recommendations, not as a source of experimentally verified research results.

3. LITERATURE REVIEW METHOD

This document uses a focused narrative-review approach. Relevant published research was located through scholarly and publisher records, including PubMed, PubMed Central and BMC/Springer Nature pages. The search focused on research about hospital information-system evaluation criteria, technical requirements and electronic-record usability. The selected publications were chosen because their topics can inform the design and evaluation of a hospital-management software project.

3.1 Selection Focus

The included studies cover three complementary questions: (a) what dimensions should be considered when evaluating an HIS, (b) which technical requirements can be checked, and (c) how usability evaluation has been conducted for hospital electronic-record systems. This is a focused selection rather than a complete systematic review of every relevant publication.

3.2 Extraction and Synthesis

For each selected study, the review records the publication details, the research purpose, the general method, the main findings reported by the authors and the implications for SmartCare. The synthesis is thematic: system-level evaluation, technical quality and usability are discussed separately and then combined into a project-oriented framework.

3.3 Interpretation Rules

Reported numerical findings are attributed to the original studies and should not be interpreted as measurements of SmartCare. Recommendations for SmartCare are reasoned applications of those findings. Where the available project information does not establish that a function has been tested, this review describes a test that should be performed rather than reporting a successful result.

3.4 Limitations of the Method

Because the review is focused and narrative, it does not claim exhaustive database coverage, duplicate independent screening, formal risk-of-bias assessment or a registered systematic-review protocol. Its purpose is to provide a structured academic foundation and an actionable quality checklist for the project. A formal systematic review would require a documented reproducible search strategy, explicit eligibility criteria and full-text screening.

4. REVIEW OF RELATED RESEARCH

4.1 Khorasani et al. (2026): Hospital Information Systems Evaluation Criteria

Khorasani and colleagues published a scoping review that aimed to identify and organise criteria used to evaluate hospital information systems. The review searched several scholarly sources and reported 52 included studies from an initial set of 561 records. Its findings grouped evaluation indicators into six broad domains: human, technological, informational, organizational, legal and patient safety. The authors also organised indicators using the structure, process and outcome perspective.

The importance of this study is its broad view of HIS quality. A system cannot be judged only by technical functionality because users, organisational procedures, information quality, legal responsibilities and patient-safety considerations also shape performance. A technically stable application may still be unsuitable if users cannot understand it or if its records are incomplete. Similarly, a usable application may be inadequate if access is not properly controlled.

Application to SmartCare: the project should be assessed across multiple dimensions. Human factors can be considered through role-based usability scenarios. Technology can be assessed through response time, reliability and deployment checks. Information quality can be checked through required fields, validation and database consistency. Organisational fit can be assessed by following realistic workflows across modules. Legal and patient-safety considerations should be treated as design constraints, while recognising that a student prototype does not itself demonstrate legal compliance or clinical safety.

The study's contribution is a comprehensive evaluation perspective rather than a ready-made test suite for SmartCare. Its categories can guide the project's evaluation checklist, but the exact test cases and acceptance thresholds must be defined for the project's scope and validated with suitable stakeholders.

Source: Khorasani, H., Damiri, S., Abbasi, S., et al. (2026). Hospital information systems evaluation criteria: a scoping review. BMC Health Services Research, 26, 184.

4.2 Relevance of the Structure–Process–Outcome View

The structure–process–outcome perspective helps separate the resources available to a system from the work performed with it and the results that follow. For SmartCare, structure may include the application, database, roles and deployment environment. Process may include patient registration, appointment handling, consultation documentation and billing. Outcomes may include fewer duplicate entries, more complete records or faster completion of a workflow. Such outcomes must be measured in a defined evaluation; they should not be assumed merely because the software has been implemented.

5. TECHNICAL REQUIREMENTS AND ARCHITECTURE

5.1 Farzandipour et al. (2020): Technical Requirements Framework

Farzandipour and colleagues developed and evaluated a framework of technical requirements for hospital information systems. The researchers used expert discussions and Delphi techniques to establish a 73-item list. The final framework covered four domains: communication service, system architecture, security service and system response time. In the hospitals assessed in their study, the reported fulfilment percentages were 63.8% for communication service, 65.5% for architecture, 72.4% for security and 76.3% for response time. These percentages describe the systems in that study, not SmartCare.

The paper illustrates why technical evaluation should be planned rather than improvised. Communication requirements concern the ability of system components to exchange information reliably. Architecture concerns how components are organised and maintained. Security requirements concern protection against unauthorised access and misuse. Response time concerns how promptly the system reacts to user actions under expected conditions.

Application to SmartCare: the Java/Jakarta Servlet/JSP and JDBC/MySQL stack should be tested for correct request handling, database connectivity, error recovery and consistency across modules. Authentication and authorization should be tested separately: logging in successfully does not prove that a user is restricted to the correct functions. Password storage should be verified to use an appropriate one-way password-hashing mechanism, and secrets should not be embedded in source code. Deployment checks should cover configuration, startup, logs and recovery from unavailable services.

5.2 Architecture and Separation of Responsibilities

A maintainable application separates presentation, application logic and data access as far as practical. JSP pages and servlet controllers can support the web interaction layer, service logic can apply business rules, and JDBC-based repositories can handle database operations. Clear separation reduces duplication and makes it easier to test rules such as appointment conflicts, billing calculations and role restrictions. This is a design recommendation for the project; the exact architecture should be confirmed against the source code.

5.3 Response Time and Reliability

A response-time test should specify the action, test data, environment, number of repetitions and observed time. Examples include opening the patient list, searching doctors, saving an appointment and generating a report. Reliability tests should include invalid input, missing database connectivity and expired sessions. Results should be recorded with the environment and limitations, because a single successful local run does not establish production reliability.

Source: Farzandipour, M., Meidani, Z., Nabovati, E., Sadeqi Jabali, M., & Dehghan Banadaki, R. (2020). Technical requirements framework of hospital information systems: design and evaluation. BMC Medical Informatics and Decision Making, 20, 61.

6. USABILITY OF HOSPITAL SOFTWARE

6.1 Wronikowska et al. (2021): Systematic Review of Usability Metrics

Wronikowska and colleagues reviewed methods and metrics used to evaluate hospital electronic health-record systems intended for clinical staff. Their search identified 1,336 records, and 51 studies were included. The authors reported 11 distinct usability-evaluation methods, seven usability metrics and 70 measurement techniques. The most frequently reported metrics included ISO 9241-11 and Nielsen's usability components; usefulness was also reported in a substantial proportion of studies. The authors concluded that usability methods and reporting varied across the field.

This review is relevant because a hospital application may contain many functions but still impose unnecessary effort on users. Usability includes whether users can complete tasks effectively, how much effort or time tasks require, and whether the system supports user satisfaction in the intended context. For a multi-role system, usability should be evaluated separately for administrators, doctors, nurses, pharmacists, laboratory personnel and billing staff because their tasks and information needs differ.

Application to SmartCare: prepare task-based scenarios instead of asking only whether users like the interface. Example tasks include registering a patient, locating a doctor, scheduling an appointment, finding a patient's appointment, entering a consultation note, recording a pharmacy transaction and preparing a bill. For each task, record completion, errors, time taken and user feedback. The project should not report usability scores unless a defined method and actual observations support them.

6.2 Practical Usability Heuristics

Useful inspection questions include: Does the system show clear feedback after saving? Are error messages understandable? Can users recognise the current patient and task? Are labels consistent between pages? Are destructive actions confirmed? Are forms arranged in a logical sequence? Can a user recover from an invalid entry without re-entering everything? Do role-specific screens avoid unnecessary options? These checks are practical recommendations informed by established usability practice; they are not findings from a new SmartCare user study.

6.3 Accessibility and Error Prevention

Readable text, sufficient contrast, keyboard access, clear focus indicators and descriptive labels can make a web interface easier to use. Required fields and validation should prevent common mistakes while explaining how to correct them. Date formats, patient identifiers and monetary values should be displayed consistently. In healthcare administration, confirmation of identity and context is especially important before editing a record or completing a transaction.

Source: Wronikowska, M. W., Malycha, J., Morgan, L. J., et al. (2021). Systematic review of applied usability metrics within usability evaluation methods for hospital electronic healthcare record systems. Journal of Evaluation in Clinical Practice, 27(6), 1403–1416.

7. USABILITY EVALUATION AND LOCAL CONTEXT

7.1 Heuristic Evaluation of Hospital Information-System Subsystems

A 2018 study titled “Usability Evaluation of Three Admission and Medical Records Subsystems Integrated into Nationwide Hospital Information Systems: Heuristic Evaluation” examined usability through a heuristic-evaluation approach. The focus on admission and medical-record subsystems is directly relevant to hospital-management applications because these workflows depend on accurate patient identification, clear navigation and consistent handling of records.

A heuristic evaluation is typically performed by inspecting an interface against recognised usability principles. It can identify issues such as unclear system status, inconsistent wording, weak error prevention or difficult navigation. It is useful early in development because reviewers can find visible interaction problems before conducting larger user studies. However, heuristic evaluation does not replace testing with representative users or verification of database and security behaviour.

Application to SmartCare: review the registration, admission, appointment and record screens for consistency. A reviewer should follow a complete task, not just inspect each page in isolation. For example, after selecting a patient, verify that subsequent pages continue to display the correct identity and that the user can tell whether a change has been saved. Record each issue with a unique ID, steps to reproduce, expected behaviour, actual behaviour, severity and evidence.

7.2 Electronic Health Records in Indian Hospitals

Research published in 2017 examined electronic health-record use in hospitals in Chandigarh, India. This topic is useful for considering adoption in the Indian hospital context, where technology use can vary according to organisational resources, staff practices, existing processes and infrastructure. The existence of an electronic system alone does not guarantee consistent adoption; implementation and workflow fit also matter.

For SmartCare, local-context considerations may include straightforward English labels, clear date and currency formats, manageable workflows for busy staff and support for realistic hospital roles. Any claim about adoption rates or local user needs should be based on the original study or direct user research, not assumed from the location alone. This review therefore uses the Indian study as contextual motivation rather than asserting specific results that are not detailed here.

7.3 Implications for Student Projects

A student project can demonstrate an understanding of usability through a documented inspection and a small, clearly described task test. It should identify the participant profile, test environment, task instructions, success criteria and limitations. If testing is performed only by the development team, that limitation should be stated because developers may overlook problems that new users encounter.

Source: “Usability Evaluation of Three Admission and Medical Records Subsystems Integrated into Nationwide Hospital Information Systems: Heuristic Evaluation” (2018), PubMed record. Indian context source: “Electronic health record use in an affluent region in India: Findings from a survey of Chandigarh hospitals” (2017), International Journal of Medical Informatics, 103, 78–82.

8. THEMATIC SYNTHESIS OF THE LITERATURE

The selected studies complement one another. Khorasani et al. (2026) provide a broad set of evaluation domains, Farzandipour et al. (2020) give a more technical checklist perspective, and Wronikowska et al. (2021) highlight the variability of usability evaluation. The 2018 heuristic-evaluation study reinforces the value of examining real admission and record interfaces. The Indian EHR study provides a reminder that implementation occurs within a specific organisational and regional context.

Table 1 summarises how these sources can inform project planning. It does not imply that the studies evaluated SmartCare.

Source	Main focus	Reported contribution	SmartCare implication
Khorasani et al. (2026)	HIS evaluation criteria	Six domains: human, technological, informational, organizational, legal and patient safety	Use a multidimensional quality checklist
Farzandipour et al. (2020)	Technical requirements	73-item framework in four technical domains	Check architecture, security, communication and response time
Wronikowska et al. (2021)	Usability metrics and methods	Identified variation in evaluation methods, metrics and measurements	Define repeatable task-based usability measures
Heuristic evaluation study (2018)	Admission and medical records interfaces	Applies heuristic evaluation to hospital subsystems	Inspect registration, admission and record workflows
Chandigarh EHR study (2017)	EHR use in Indian hospitals	Examines EHR use in a regional hospital context	Consider adoption and local workflow fit; verify details from full paper

8.1 Common Themes

Four common themes emerge. First, quality must be evaluated across several dimensions rather than by counting features. Second, technical architecture, security and response time require explicit verification. Third, usability should be measured through user tasks and structured observations. Fourth, implementation context matters: the system should fit the roles and processes of the organisation in which it is used.

8.2 Research Gap for SmartCare

The reviewed literature does not provide empirical evidence about SmartCare itself. A project-specific evaluation could address this gap by documenting test cases, recording results, measuring a small set of user tasks and clearly describing the test environment. Such evidence would make the project report more credible than unsupported statements that the system is fast, secure or easy to use.

9. APPLICATION TO SMARTCARE

SmartCare's described modules span administrative and operational workflows. The following mapping translates literature themes into project-level checks. These are proposed checks and should be marked complete only after they have been executed and documented.

9.1 Authentication and Administration

Verify that valid users can sign in, invalid credentials are rejected, logout ends the session and protected pages cannot be opened without appropriate authentication. Test role restrictions by attempting both permitted and prohibited actions. Confirm that password storage uses the intended BCrypt-based approach and that passwords are never displayed in logs or error messages. Confirm that configuration secrets are stored outside public source files.

9.2 Patient and Doctor Management

Check required fields, input formats, duplicate handling, search behaviour and update consistency. Verify that a patient can be found using the expected identifier and that editing one record does not change another. Doctor records should be associated with the appropriate speciality or availability information if those fields are part of the implementation. Test empty results and invalid search terms.

9.3 Appointments and Consultation

Test creation, rescheduling and cancellation of appointments, including missing patients, unavailable doctors and conflicting time slots if conflict prevention is implemented. Confirm that the selected patient and doctor remain consistent through the workflow. Consultation data should be linked to the correct encounter and should not be visible to users without appropriate permissions.

9.4 Pharmacy, Laboratory and Nursing

For pharmacy workflows, validate medicine details, quantity, price and any implemented stock rules. For laboratory workflows, check order identifiers, result status and association with the correct patient. For nursing workflows, check the accuracy and traceability of recorded observations or care-related entries. The exact scope depends on the functions actually implemented; unsupported features should be documented as out of scope rather than presented as complete.

9.5 Billing, Admissions, Reports and Feedback

Check arithmetic and rounding in billing, relationships between bills and services, admission and discharge state changes, report filters and feedback submission. Reports should use consistent date ranges and avoid exposing information to unauthorised roles. Feedback should show a clear confirmation after successful submission and a meaningful message if submission fails.

10. PROPOSED TESTING AND EVALUATION FRAMEWORK

The literature suggests that SmartCare should be evaluated through a combination of functional, technical, security and usability checks. The proposed framework below is a plan, not a record of completed tests.

10.1 Functional Testing

Functional tests verify that each feature follows its defined requirements. For every test, record a test ID, module, preconditions, steps, expected result, actual result, status, date and evidence. Include positive cases, negative cases, boundary values and cross-module cases. Example: create a patient, schedule an appointment for that patient, then confirm that the appointment references the correct patient and doctor.

10.2 Integration Testing

Integration tests examine whether modules work correctly together. Focus on the data passed between registration, appointments, consultation, pharmacy, laboratory, billing and reports. Check that updates appear where expected, that invalid relationships are rejected and that partial failures do not leave misleading or inconsistent records. If a transaction updates multiple related tables, verify whether the implementation preserves consistency when one operation fails.

10.3 Security Testing

Security tests should cover authentication, authorization, session management, input validation, database access and error handling. Verify that SQL queries use parameterised statements where applicable and that error responses do not expose credentials, stack traces or sensitive patient information. Use only synthetic test data for academic testing. A checklist or student test does not replace a professional security assessment, privacy review or compliance audit.

10.4 Performance and Reliability

Choose representative tasks and define a repeatable environment. Record response time over multiple runs, the test data size, concurrent-user assumptions and any failures. Test the behaviour when the database is unavailable, a form contains invalid values or a session expires. A response-time target should be set before testing rather than selected after observing the result.

10.5 Usability Testing

Ask representative users to complete a small set of tasks without step-by-step assistance. Record task completion, errors, time and comments. Avoid collecting real patient information. After the tasks, ask which labels or steps were confusing and which information was difficult to locate. If only classmates or developers participate, clearly identify that limitation and avoid generalising to all hospital staff.

10.6 Defect Management

Each issue should have a unique identifier, concise title, severity, priority, environment, reproduction steps, expected and actual behaviour, supporting evidence and resolution status. Retest fixes and check for regressions in related modules. This produces a traceable quality record that can be included in the final project documentation.

11. PROPOSED METRICS AND TEST MATRIX

Metrics turn broad quality goals into observable evidence. They should be selected before execution and interpreted within the limits of the test environment.

Quality area	Example measure	Evidence to record
Functional correctness	Passed tests / executed tests	Test-case sheet and screenshots/logs
Data integrity	Correct relationships and updates	Database checks and cross-module scenarios
Usability	Task completion, errors, time	Observation sheet and participant feedback
Performance	Median or average response time	Environment, repetitions and timings
Security	Authorisation and session test outcomes	Role matrix and negative test cases
Reliability	Recovery and error-handling behaviour	Failure scenario and recovery record
Maintainability	Build reproducibility and defect traceability	Build log, configuration notes and issue list

11.1 Suggested Acceptance Criteria

Acceptance criteria should be based on the requirements agreed for the project. For example, all critical authentication and access-control tests should pass before demonstrating protected functions; patient and appointment relationships should remain consistent; invalid form submissions should produce understandable errors; and core workflows should not silently lose data. Performance criteria should specify a threshold and test environment. If a criterion has not been tested, label it “not evaluated” rather than “passed.”

11.2 Sample Test Scenario

Scenario: Appointment creation for a registered patient.
Precondition: A synthetic patient and an available doctor exist in the test database.
Steps: Sign in with an authorised role; open appointments; select the patient and doctor; enter a valid date and time; save; then search for the appointment.
Expected result: The appointment is saved once, linked to the selected patient and doctor, and shown in the appointment list. Invalid or conflicting details should be handled according to the requirements.
Evidence: Test ID, input data, result, screenshot or log, and tester/date.

11.3 Reporting Results Responsibly

A project report should distinguish completed testing from planned testing and distinguish observations from conclusions. If the live deployment is unavailable during a test session, record that limitation and repeat the test when the environment is accessible. Do not infer that a module works simply because the source code contains a corresponding page or class.

12. DISCUSSION, LIMITATIONS AND FUTURE WORK

12.1 Discussion

The literature supports a balanced view of hospital software quality. The broad evaluation domains in Khorasani et al. (2026) help prevent an overly narrow focus on code or interface appearance. The technical framework from Farzandipour et al. (2020) provides concrete areas for verification, while Wronikowska et al. (2021) show that usability evaluation can vary substantially unless methods and measures are specified. Taken together, these studies support a repeatable, evidence-based evaluation process.

For SmartCare, the key engineering challenge is not simply to provide many modules; it is to ensure that data and workflow transitions are consistent, access is appropriate and users can complete tasks clearly. A modular architecture can help maintainability, but architecture alone does not prove quality. Security mechanisms such as password hashing are important, but they do not replace role-based authorisation, safe session handling, secure configuration and testing. Similarly, a visually clear screen does not prove that its data relationships are correct.

12.2 Limitations

This review has several limitations. It uses a focused selection of studies rather than a comprehensive systematic search. Some sources are represented through their indexed records and abstracts, so detailed methodological or numerical claims beyond those records have not been included. The review does not conduct a new experiment, interview hospital staff, or measure SmartCare's performance. It also does not establish clinical effectiveness, regulatory compliance, privacy compliance or production readiness.

The live project environment was not independently verified as part of this document. Consequently, the report does not claim successful end-to-end testing of the deployed application. The module mapping is based on the project description and should be reconciled with the final source code and implemented features before submission.

12.3 Future Work

Future work could include: (1) a requirements traceability matrix linking each feature to test cases; (2) structured user testing with representative roles and synthetic data; (3) database integrity and concurrency testing; (4) a security review of authentication, permissions, sessions and configuration; (5) repeatable performance tests under defined loads; (6) accessibility inspection; and (7) deployment documentation covering backup, restore, monitoring and incident handling. Any clinical deployment would require substantially more governance and validation than a classroom demonstration.

12.4 Ethical and Privacy Considerations

Hospital records can contain highly sensitive personal information. Development and testing should use fictional or properly anonymised records, restrict access to test environments and avoid exposing data in screenshots, logs or public repositories. The project should not be used to make real clinical decisions. If future development involves real patient data, the team must obtain appropriate institutional guidance and follow applicable privacy, security and ethical requirements.

13. CONCLUSION

This research review examined published work on hospital information-system evaluation, technical requirements and electronic-record usability, then applied those themes to SmartCare. The reviewed studies show that HIS quality is multidimensional: technology, information quality, people, organisational processes, security, usability and patient-safety considerations all matter. Technical frameworks help define what to test, while usability methods help determine whether intended users can complete important tasks effectively.

For SmartCare, the practical recommendation is to evaluate the system through documented functional, integration, security, performance, reliability and usability tests. Each result should be traceable to a test case and supported by evidence. Features that have not been tested should be labelled accordingly. This approach strengthens the project academically and avoids unsupported claims about security, speed or clinical readiness.

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